

A Four-Layer Model of Student Struggle in IGCSE/IAL: A Conceptual Framework for the Bangladesh English-Medium Context

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Abstract

Some students finish their IGCSE and A-Level examinations in a single sitting, without re-takes, with straight A*s. Others retake the same paper three, four, five times, studying obsessively between each attempt, and still cannot cross the grade boundary. In Bangladesh's rapidly growing English-medium sector, where these qualifications gatekeep access to overseas university admission, this bimodal distribution of outcomes is visible in every classroom and every coaching centre, yet remains unexplained in the academic literature. This paper asks a simple question: what distinguishes the students who pass everything in one go from those who get trapped in the retake cycle, and why does studying harder not help?

Drawing on my dual position as a self-taught private-candidate student who compressed six O-Level subjects into six months of self-study and as a current tutor in Dhaka, I synthesise the educational-psychology and cognitive-science literatures with the Bangladesh English-medium context to propose a four-layer model of academic performance: Content Knowledge $\times$ Exam Literacy $\times$ Learning Strategy $\times$ Self-Regulation. The model explains the effort paradox: why some students achieve top grades with apparent minimum effort while others study 200 hours and fail, as a consequence of differential layer strength rather than differential intelligence. I develop a six-type typology of student struggles (prerequisite gap, surface engagement, exam literacy, metacognitive, motivational, and instructional-environment), each with a hypothesised behavioural signature testable through interviews, school records, or learning-platform data, and I articulate a mechanism for the retake paradox in which failed retakes compound self-efficacy erosion into self-reinforcing cycles. The paper pre-registers six testable predictions about the differences between one-sitting completers and chronic retakers. A companion paper (Khondokar, 2026) describes an AI-augmented learning management system designed to detect and address each struggle type at scale. This paper is a conceptual proposal and pre-registration, not an empirical study with data; the predictions are hypotheses to be tested in subsequent cycles.

Keywords: IGCSE, International A-Level, Edexcel, Cambridge, Bangladesh, English-medium education, self-regulated learning, cognitive load theory, mastery learning, exam

1 Introduction

1.1 Background and Positionality

This paper is written from a dual position: as a former private-candidate student who completed the Edexcel IGCSE and International A-Level qualifications through self-study rather than the conventional school-and-coaching-centre pathway available to better-resourced families, and as a current tutor working with IGCSE/IAL students in Dhaka across the ability spectrum. The paper’s attention to private candidacy, coaching-centre economics, school-tier stratification, and exam literacy reflects lived proximity to these mechanisms as a middle-class, financially constrained participant in the system, rather than solely a literature-derived interest in them. This positionality is disclosed because it shapes the analysis throughout: I did not have the option of pursuing the full school-plus-coaching-plus-private-tutor pathway available to wealthier families, and found, in private candidacy and self-study, the route that an increasing number of middle-class Bangladesh English-medium students are pushed toward as the cost of formal schooling outpaces household incomes. Consistent with the anonymised review process for this submission, further autobiographical detail is withheld from the manuscript body.

The Bangladesh English-medium sector itself comprises a privately financed strand of education that follows international curricula, predominantly the Edexcel and Cambridge IGCSE and International A-Level qualifications. The sector has grown rapidly since the 1990s, fuelled by demand from middle- and upper-middle-class families who view these qualifications as routes to overseas university admission and as markers of social distinction (Chowdhury and Kabir, 2010; Hamid, 2009). Although reliable national statistics are scarce, the British Council and the examination boards report year-on-year increases in entries from Bangladesh, and the sector now serves tens of thousands of candidates annually across more than two hundred registered centres (Kader and Salam, 2018). I am both a product of this system and a current teacher within it, and this paper grows out of that friction, observed from both positions over the past decade.

Within this population, achievement outcomes display a markedly bimodal distribution. A subset of students complete IGCSE and IAL in a single sitting, often achieving A or A* grades across multiple subjects, and proceed to university admission without retakes. A second, non-trivial subset enters prolonged retake cycles, repeating the same paper across multiple examination sessions over two or more years and frequently failing to cross the grade boundary required by their target university. Anecdotally, teachers in the sector attribute the divergence to “discipline” or “talent”, but these folk explanations are underspecified and offer no actionable intervention points. Public discourse on Reddit ([r/igcse](#), [r/alevel](#), [r/Olevels](#), [r/alevels](#), [r/6thForm](#), [r/Dhaka](#), [r/bangladesh](#), [r/Edexcel](#)), The Student Room, Quora, and Bangladesh English-medium Facebook groups reveals the same bimodal pattern, often in stark first-person terms: students report studying 75 consecutive days and failing, retaking the same paper for the third or fifth time, internalising parental disappointment, and questioning whether they are simply “not smart enough”.

The socioeconomic dimension of this bimodality is sharp. There is a recent trend in which students, after completing O-Levels, skip A-Levels entirely and instead do a foundation

course, typically in the UK, Australia, or Malaysia, that leads directly into an undergraduate programme. This route is expensive, foundation courses plus one year of overseas living can cost €20,000 to €40,000, but it bypasses the A-Level bottleneck entirely. Rich students can take this escape hatch. Middle-class students cannot. For middle-class students, the only route into a reputed Bangladeshi university, or into a reasonably-ranked overseas university with scholarship, is to complete A-Levels with strong grades. This means that the retake paradox, the coaching-centre dependency, and the teacher-quality bottleneck fall disproportionately on middle-class students. Rich students buy their way out of the system; poor students never enter it. The middle class absorbs the structural failure.

1.2 Problem Statement

The gap in the literature has practical consequences. Private tutors and coaching centres in Bangladesh operate on a model that is largely reactive: students are drilled on past papers in the weeks preceding an examination, with little diagnostic attention to prerequisite knowledge gaps or to the cognitive processes that examination questions are designed to elicit. Where a student fails a paper, the standard response is more drilling, not diagnostic remediation (Hossain et al., 2025). This pattern is particularly striking because the IGCSE and IAL syllabi are public, well-structured documents, and because official mark schemes, examiner reports, and grade boundaries are all freely available. The information required for diagnostic teaching exists; what is missing is the conceptual framework to use it at scale and at the level of individual students.

The human cost of this missing framework is not abstract. In my practitioner network, students who exhaust several retake cycles without crossing the grade boundary commonly abandon the qualification and enter the workforce rather than continuing. Because the universities that accept A-Level retakers are few, and the ones that do require grades that repeated retakers rarely achieve, this route effectively excludes them from Bangladeshi higher education. Had they been in the Bangla-medium system, the higher pass rates, grade inflation, and larger university intake of the SSC/HSC pathway would likely have absorbed them; the smaller, less-inflated English-medium system does not. The retake dead-end in Bangladesh English-medium education is therefore not only an academic problem but a life-trajectory one, and the system currently offers no soft landing for students who fall through it.

1.3 Research Questions

This paper is organised around two research questions:

1. **RQ1.** What are the primary systemic, pedagogical, psychological, and cognitive sources of struggle for Edexcel and Cambridge IGCSE/IAL students in Bangladesh, and what factors distinguish students who complete the qualification in a single sitting from those who enter prolonged retake cycles?
2. **RQ2.** How can the constructs of self-regulated learning, deliberate practice, cognitive engagement, cognitive load, and exam literacy be operationalised as observable behavioural and outcome signatures, testable by any researcher with access to interviews, school records, or a generic learning platform?

1.4 Significance

This paper makes three contributions. First, it synthesises literature that has not previously been brought together: the Bangladesh English-medium literature, which is largely sociolinguistic, and the educational psychology and cognitive science literatures, which are largely Anglo-American. Second, it proposes a four-layer model of academic performance and a six-type typology of student struggles, each with a hypothesised observable signature. Third, it pre-registers six testable predictions about behavioural and outcome differences between one-sitting completers and chronic retakers.

A companion paper (Khondokar, 2026) describes a learning management system, currently under development, that is designed to detect and address each struggle type at scale through a knowledge-graph-grounded retrieval-augmented generation architecture. The present paper is legible without any knowledge of that system: the typology and predictions are framed in terms observable through interviews, school records, or any sufficiently instrumented learning platform. The companion system is one possible operationalisation of the framework; it is not a prerequisite for the framework itself.

1.5 Companion Paper and Division of Labour

This paper is submitted as a companion pair with Khondokar (2026), which describes the SyllabAI system. To clarify the division of labour between the two papers and to forestall ambiguity for reviewers and readers:

- **What this paper (Paper A) claims.** The conceptual framework, the four-layer model, the six-type struggle typology, and the six pre-registered predictions. All claims in this paper are framed in terms observable through any sufficiently instrumented research setting (interviews, school records, or a generic learning platform).
- **What this paper defers.** Empirical validation of the predictions; the design, deployment, and evaluation of any particular system; the splits of Types 3 and 5 into 3a/3b and 5a/5b sub-types (recommended by a separate analytic autoethnography, see Section 6). The splits are treated in this paper as hypotheses to be tested in Cycle 1 rather than as adopted structure.
- **What the companion paper (Paper B) claims.** The SyllabAI system architecture (knowledge graph, proficiency overlay, KA-RAG chatbot, learning log) and the pre-registration of system-specific predictions about telemetry signatures, mediated effects, LLM marking reliability, diagnostic-overhead reduction, and cross-topic question detection.
- **Which Paper A predictions are testable without the SyllabAI system.** Predictions 1–6 (Section 5) are designed to be testable through interviews, school records, or any learning platform. They do not require SyllabAI. Where SyllabAI telemetry is available, the predictions can be tested at higher statistical power; where it is not, the predictions remain testable through other instruments.

The two papers should remain separate publications because they make distinct contributions (conceptual vs. system-design) to distinct scholarly communities (educational psy-

chology vs. learning sciences). Cross-references are kept minimal in this paper to preserve its legibility without the companion.

The remainder of this paper is structured as follows. Section 2 reviews the relevant literature. Section 3 presents the conceptual framework, the four-layer model, the K.E.A.R.N. Gap, and the typology of struggles. Section 4 describes the mixed-methods design-based research programme within which the typology will be validated. Section 5 pre-registers six testable predictions. Section 7 concludes.

2 Literature Review

2.1 English-Medium Education in Bangladesh

The English-medium sector in Bangladesh occupies an unusual position. Unlike state-sector schools, which follow the national curriculum in Bangla, English-medium schools follow international curricula taught in English, and they prepare students for IGCSE and IAL examinations administered by Pearson Edexcel and Cambridge International. The sector has been studied primarily through the lens of language policy and identity. Hamid (2009) examined the rural–urban divide in English language teaching and noted the symbolic capital attached to English-medium education in urban Bangladesh. Chowdhury and Kabir (2010) analysed the “language wars” that frame English-medium education as a site of class distinction, where access to English is treated as a marker of cosmopolitan identity. Hamid et al. (2014) extended this analysis across Asia, observing that medium-of-instruction policies in the region are typically driven by globalisation pressures rather than by pedagogical evidence.

Empirical work on the Bangladesh English-medium sector itself has begun to accumulate. Kader and Salam (2018) conducted a comprehensive study of service quality and satisfaction levels in the English-medium education system, surveying students, guardians, and teachers. The study identified significant variations in institutional quality, with respondents reporting concerns about the lack of standardised resources, qualified teaching personnel, and consistent assessment practices. Hossain et al. (2025) examined the related phenomenon of English-version education, where the national curriculum is delivered in English, identifying problems including a lack of trained teachers capable of delivering content in English, inadequate learning materials aligned with international standards, and a disconnect between curriculum expectations and students’ English language proficiency. Together, these studies paint a picture of a sector that has expanded faster than its supporting infrastructure, with school quality varying dramatically across institutions.

A literature search conducted for this revision (targeted web search, July 2026, for terms including “IGCSE/IAL Bangladesh”, “English-medium retake Bangladesh”, “private candidate Bangladesh English-medium”, and “foundation course Bangladesh A-Level”; this was a targeted check rather than a systematic review) located no post-2022 peer-reviewed empirical study that directly tests the within-population outcome variance that this paper addresses. Post-COVID empirical work on Bangladeshi education exists (e.g., systematic reviews of online-learning disruption in Bangladeshi schools and universities), but it has focused on school-level online-learning disruption, primary-school dropout, or higher-education teacher practice rather than on IGCSE/IAL retake patterns or on the bimodal outcome distribution within the English-medium sector. The gap claim on which this paper rests, namely that no published study has

tested whether self-regulated-learning, cognitive-load, and exam-literacy constructs predict the bimodal IGCSE/IAL outcome distribution in Bangladesh, therefore stands as stated in the original submission. A more thorough systematic review is planned as part of the Cycle 1 protocol (Section 4); the framework proposed here is intended to provide a diagnostic vocabulary that future empirical work can use to disaggregate the chronic-retaker population by struggle type, and the first research cycle is designed to begin filling this gap directly.

The school market stratification observed in this practitioner context maps cleanly onto this empirical picture. At the top end are schools such as Scholastica and DPS, which charge high fees, hire teachers with strong higher-education credentials and good English proficiency, and produce students who routinely achieve A and A* grades. Even at these schools, however, students commonly attend coaching centres in the evenings, which reveals the limits of even the best Bangladesh English-medium schooling. In the middle tier are schools such as Don Bosco, which charge lower fees, still three to five times a Bangla-medium school, but hire teachers who are, in many cases, Bangla-medium graduates themselves. These teachers did not sit IGCSE or IAL examinations as students; they learned the syllabus as adults, often by working through the textbook the year before they teach it. Their command of the mark scheme, command words, and examiner expectations is correspondingly weak. Students at these schools depend heavily on coaching centres to compensate. At the lower tier are the cheapest English-medium schools available, still more expensive than Bangla-medium or English-version schools, serving families who want the English-medium credential but cannot afford the top tier. Teacher quality at this tier is poor, individual attention to weak students is largely absent, and the school's business model depends on enrolment volume rather than academic outcomes. The critical observation is that the price signal in the Bangladesh English-medium market is not perfectly correlated with teacher quality, but it is strongly correlated with the school's ability to attract teachers who themselves sat IGCSE/IAL or who have subject-specialist training.

The coaching-centre economy has, over the past decade, hollowed out the senior sections of these schools. As the coaching-centre system expanded, students who initially attended coaching centres for supplementary help began to leave school entirely and study only at coaching centres. Senior sections emptied, and schools discontinued them rather than invest in filling them. This pattern, schools hollowing out at the senior level as students migrate to coaching centres, is now the de facto standard in Bangladesh English-medium education. Schools have responded by creating their own internal coaching batches: regular teachers take after-school classes for a separate fee, blurring the line between school and coaching centre. The school day becomes a loss leader; the real instruction happens in the after-school batch. The implication is that the formal school system in Bangladesh English-medium has, at the senior level, largely ceased to be the primary site of instruction. The coaching centre has taken over, and the school functions as an administrative shell that handles exam registration and, in some cases, predicted grades.

This coaching-centre phenomenon is not a Bangladesh-specific anomaly. It is a documented instance of what the comparative education literature calls "shadow education" that is, private supplementary tutoring that shadows the mainstream school system in curriculum, timing, and content (Bray, 1999). The UNESCO International Institute for Educational Planning has monitored shadow education globally since the late 1990s, and the Asia-Pacific region shows the highest prevalence rates worldwide, with some countries reporting that over 90% of secondary students receive or have received private tutoring. The Bangladesh coaching-centre system fits the global pattern: it expands when mainstream schooling is perceived as inadequate, it is fuelled by middle-class anxiety about competitive examinations, and it creates

a parallel instructional system that the formal school system eventually accommodates rather than competes with. Alam and Zhu (2022) document the Bangladesh-specific trajectory of this expansion, noting that low teacher salaries in the mainstream system actively encourage teachers to supplement their income through private tutoring and coaching classes, which further entrenches the shadow system. Framing the Bangladesh coaching centres as shadow education connects the local observations in this paper to a substantial body of comparative research on the phenomenon.

One feature of the English-medium system that distinguishes it from Bangla-medium is the ability to register as a private candidate. Students do not have to be enrolled in a school to sit IGCSE or IAL examinations. This creates a survival route for middle-class families who cannot afford the full school-plus-coaching-plus-private-tutor stack. The common pattern is that middle-class English-medium students leave school around Class 9 or 10, register privately for the exams, and study at coaching centres or on their own. Rich students stay in school, for the social capital and the predicted-grade credibility, and also attend coaching centres. Poor students cannot enter the English-medium track at all. Private candidacy functions as the middle-class compromise: a route that allows families to avoid paying for instruction they cannot afford.

A further institutional feature that bears on the validity of cross-system comparisons in Bangladesh is the integrity of the examination process itself. In IGCSE and IAL exam halls, cheating is rare: invigilation is strict, the consequences are severe, and the cultural expectation among English-medium students is that the exam measures their own work, with typically more than fifty invigilators in an IGCSE/IAL hall, two per row. By contrast, normalised cheating during examinations is a widely reported feature of the Bangla-medium tradition, where a single invigilator per hall is common and collusion is widely tolerated. A recent viral social-media post from a Bangla-medium student sitting her SSC exam described her shock at finding three invigilators in the hall, which by her standards was unusually many, and she described the environment as hostile on those grounds, asking how students were expected to sit the exam under such scrutiny. The point here is not to moralise about cheating but to note that institutional integrity in the examination process is a precondition for the validity of the credential it produces. When the credential is inflated by cheating, it ceases to reliably signal competence, and downstream selection processes, university admission, job markets, become noisier. The English-medium system in Bangladesh, whatever its other problems, has maintained comparatively high institutional integrity in this respect; the research literature on Bangladesh education has not, to my knowledge, adequately addressed this structural difference.

A common belief in the Bangladesh English-medium community holds that Edexcel is “easier” than Cambridge. This is partially true and partially false, and the asymmetry matters. At IGCSE, Edexcel is generally easier than Cambridge. Edexcel IGCSE subjects have slightly less content, the questions are more direct, and the grade boundaries are more forgiving. A student choosing between Edexcel and Cambridge at O-Level should, all else equal, choose Edexcel. At A-Level, the pattern reverses. Edexcel IAL is substantially harder than Cambridge A-Level. The Edexcel IAL papers are more mathematically demanding, the questions are less predictable, and the grade boundaries are less forgiving. Cambridge A-Level, while still a significant step up from IGCSE, is more tractable. There is, however, a structural advantage to Edexcel IAL that Cambridge A-Level lacks: modularity. In Edexcel IAL, if a student fails one unit, they can retake just that unit. In Cambridge A-Level, a student who failed historically had to retake all units. This means that Edexcel IAL, despite being harder per-unit, is more forgiving overall because the retake surface is smaller. This Edexcel-versus-Cambridge asymmetry is

well-known in the Bangladesh coaching-centre community but is not, to my knowledge, documented in the academic literature. It is relevant to this research because it complicates any simple comparison of pass rates or retake patterns between the two boards. A higher Edexcel IAL retake rate may reflect the lower per-unit cost of retaking, a structural feature, rather than harder content, a pedagogical feature, or it may reflect both. Disentangling these requires board-level data.

2.2 Teacher Quality and the Instructional Environment

The teacher-quality bottleneck is the structural constraint behind the hollowing-out described above. There is a severe shortage of good English-medium teachers in Bangladesh. The shortage is structural: English-medium schooling has expanded faster than the pipeline of teachers who themselves sat IGCSE/IAL and who understand the mark schemes, command words, and examiner expectations at a deep level. The problem is compounded by burnout-driven attrition: international school teachers face systemic workload and wellbeing pressures that drive turnover (Bunnell and Burnham, 2023), and ISC Research (2024) report significant recruitment and retention challenges across the international school sector globally, creating a vicious cycle in which poor teacher retention worsens the shortage, which increases class sizes and workload for remaining teachers, which drives further attrition. Teachers new to international schools face additional challenges of cultural adaptation, language barriers, and unfamiliar curricula (Halicioğlu, 2015), which further reduces instructional quality during the transition period.

The shortage is also a problem of teacher efficacy, not just teacher knowledge: Tschannen-Moran et al. (1998) show that a teacher's belief in their own ability to help students learn predicts student outcomes independently of the teacher's actual subject knowledge. Under-qualified teachers who lack confidence in their ability to teach the syllabus are less likely to invest the diagnostic effort that struggling students need, even when they know the content. In my practitioner experience, students frequently arrive during exam season having been taught by teachers who did not understand the syllabus, who taught the wrong syllabus, or who taught content but not exam technique. Public-discourse evidence from The Student Room confirms this pattern, with students reporting teachers who skip entire topics, teach incorrect content, or fail to cover the syllabus before the examination. Tutors with genuine depth of knowledge are rare enough that the market treats them as scarce resources: demand routinely exceeds supply during peak exam season, and guardians attempt to secure exclusive access to tutors they trust.

The coaching-centre market responds to this shortage in a predictable way: it hires anyone. Many coaching centres hire teachers who scored C or D in their own IGCSE/IAL. These are not bad grades in absolute terms; a student with C or D in IGCSE, had they sat the Bangla-medium or English-version equivalent, might have been a top scorer. But they do not have the depth of mark-scheme and examiner-expectation knowledge needed to teach the subject well. The result is a market for lemons (Akerlof, 1970): students cannot reliably distinguish good coaching centres from bad ones *ex ante*, and the bad ones drive the good ones out by undercutting on price. The shadow education system thus expands as a symptom of instructional failure in the mainstream system (Alam and Zhu, 2022): low teacher salaries drive teachers into private tutoring, which reduces their investment in school teaching, which drives students into coaching centres, which reduces school revenue and teacher conditions further. The cycle is self-reinforcing, and any intervention that targets only the student without

addressing the instructional environment will be incomplete.

2.3 Self-Regulated Learning and Academic Achievement

The question of why some students succeed while others struggle, despite access to the same educational resources, has been the subject of extensive research in educational psychology. Zimmerman (2002) defines self-regulated learners as those who metacognitively, motivationally, and behaviourally participate in their own learning, monitoring their understanding, adjusting their strategies, and persisting in the face of difficulty. Pintrich (2000) distinguishes between mastery goal orientations, in which the student seeks to develop competence, and performance goal orientations, in which the student seeks to demonstrate competence relative to others. The two orientations predict different learning behaviours: mastery-oriented students tend to select challenging tasks, attribute failure to insufficient effort, and use deeper processing strategies, whereas performance-oriented students tend to select easier tasks, attribute failure to lack of ability, and use superficial strategies.

The SRL tradition comprises several overlapping models. Zimmerman (2002) proposes a three-phase cyclical model comprising forethought (goal-setting and strategy selection), performance (execution and self-monitoring), and self-reflection (evaluation and strategy adaptation) and maps directly onto the revision cycle a student navigates during exam preparation. Pintrich (2000) proposes a complementary four-phase framework (planning, monitoring, control, evaluation) that is more granular in its treatment of the monitoring and control processes. Both models predict that students who cycle through all phases effectively will outperform students who skip phases, even when total study time is held constant. A meta-analysis by Ergen and Kanadlı (2017) found that self-regulated learning (SRL) strategies had a large effect size ($d = 0.859$) on academic achievement, confirming that the capacity to plan, monitor, and evaluate one's own learning processes is among the strongest predictors of academic success. The Education Endowment Foundation's guidance report on metacognition and self-regulated learning (Education Endowment Foundation, 2018) further established that metacognitive strategies are effective across age groups and subject domains, with particular relevance for examination preparation contexts where students must independently manage their revision timelines, identify knowledge gaps, and select appropriate study strategies. The Vanderbilt Peabody IRIS Center characterises the differences between successful and struggling learners across four executive function processes: goal-setting, self-monitoring, self-evaluation, and resource management (Vanderbilt Peabody IRIS Center, 2020). Successful learners set specific, challenging goals; monitor their comprehension in real-time; evaluate their performance against external standards; and actively seek out resources when they encounter obstacles. Struggling learners, by contrast, tend to set vague goals, fail to notice comprehension failures, avoid self-evaluation, and rely passively on whatever resources are immediately available.

2.4 Cognitive Engagement, Mindset, and Self-Efficacy

Three further lines of work in educational psychology refine the picture. Chi and Wylie (2014) propose the ICAP framework, which distinguishes four levels of cognitive engagement: Interactive, Constructive, Active, and Passive. Passive engagement, receiving information without transformation, produces the weakest learning outcomes; active engagement, manipulating information without generating new content, is moderately effective; constructive engagement,

generating new content, is more effective; and interactive engagement, co-constructing with another agent, is most effective. The ICAP framework offers a useful diagnostic vocabulary for distinguishing revision behaviours: a student who reads a textbook and highlights passages is engaged passively; a student who completes past paper questions and self-marks against the mark scheme is engaged actively; a student who attempts a question, fails, and then constructs an explanation of why the mark scheme awards marks in a particular way is engaged constructively.

Dweck (2006) distinguishes between a fixed mindset, in which intelligence is viewed as a static trait, and a growth mindset, in which intelligence is viewed as malleable through effort. Mindset predicts response to failure: students with a growth mindset are more likely to interpret examination failure as a signal to adjust strategy, whereas students with a fixed mindset are more likely to interpret it as evidence of innate inadequacy. The relevance to the retake paradox is immediate: a student with a fixed mindset who fails an IGCSE paper may attribute the failure to inherent inability, repeat the same revision strategy, and fail again. The retake cycle, in this account, is not a failure of effort but a failure of metacognition.

Bandura (1997) defines self-efficacy as a person's belief in their ability to succeed in specific situations, and identifies it as a powerful predictor of academic outcomes. Students with high self-efficacy are more likely to attempt challenging problems, persist through difficulty, and recover from failure. Self-efficacy is both cause and consequence of academic performance: success builds self-efficacy, which leads to more effective study behaviours, which leads to more success; conversely, repeated failure destroys self-efficacy, leading to reduced effort, avoidance, and further failure. This creates a self-reinforcing positive feedback loop for struggling students and a virtuous cycle for successful students. Over time, these cycles create the appearance that some students are “naturally gifted” while others are “naturally bad at school” when in reality the divergence is driven by self-reinforcing behavioural patterns.

2.5 Mastery Learning, Formative Assessment, and Feedback

The argument above connects to a longer tradition of work on mastery learning and formative assessment. Bloom (1968) argued that most students can achieve mastery of a well-specified learning objective given appropriate instruction and sufficient time, and that the variance in achievement observed under conventional instruction reflects variance in prerequisite knowledge and instructional fit rather than variance in ability. Bloom's “2 sigma problem” demonstrated that one-on-one mastery tutoring produces a two-standard-deviation improvement in achievement, moving an average student to the 98th percentile. The pedagogical implication is that instruction should be diagnostic, identifying prerequisite gaps and remediating them before progressing to dependent topics.

The formative assessment tradition develops this argument. Black and Wiliam (1998) synthesised evidence that formative assessment, broadly construed as feedback that is used to adjust teaching and learning, produces substantial learning gains, frequently larger than those produced by other educational interventions. Hattie and Timperley (2007) extended this synthesis and identified the specific features of effective feedback: it must focus on the task rather than the self, it must be specific rather than general, and it must direct the student towards improved strategies rather than simply reporting correctness. Biggs (1996) framed the same point in terms of constructive alignment: assessment, teaching, and intended learning outcomes must be aligned such that the activities students undertake to prepare for assessment are the activities

that develop the intended understanding.

The relevance to IGCSE and IAL is direct. The official mark schemes published by Edexcel and Cambridge are, in effect, detailed formative assessment rubrics. They specify the cognitive operations that earn marks, the keywords and steps that examiners look for, and the common errors that lose marks. Yet in the typical Bangladesh coaching-centre context, mark schemes are used summatively rather than formatively: students complete past papers, mark themselves against the scheme, and move on. The diagnostic potential of the mark scheme is largely wasted. Cambridge International's own blog confirms this pattern, noting that "high achieving pupils often practise with past papers. Less successful students do not look at previous papers, or they might flick through the content without attempting the questions" (Cambridge International Education, 2023).

A closely related literature on assessment literacy (Popham, 1999; Stiggins, 2017) develops the parallel argument that the ability to read, interpret, and act on assessment evidence is a distinct competence that must be taught, not assumed. Where the formative-assessment tradition focuses on the teacher's use of assessment evidence, the assessment-literacy tradition focuses on the student's ability to decode what a mark scheme rewards, what a command word demands, and what a particular marking-point loss reveals about the underlying cognitive gap. Layer 2 of the four-layer model proposed in this paper (Exam Literacy) draws on this assessment-literacy tradition but narrows it to the specific case of high-stakes public examinations: the relevant "literacy" is not general assessment literacy but the specific mark-scheme vocabulary, command-word decoding, and answer-structure conventions of Edexcel and Cambridge IGCSE/IAL. The Type 3 struggle in the typology below is the failure mode that this narrower literacy addresses.

2.6 Deliberate Practice, Spacing, and the Testing Effect

A fourth line of work, drawn from the expertise literature, further specifies what "different revision" should look like. Ericsson (2006) distinguishes deliberate practice from mere repetition: deliberate practice is structured, effortful, focused on specific weaknesses, and accompanied by immediate feedback. Simply solving more past paper questions is not deliberate practice unless the student identifies which sub-skills are failing and targets those sub-skills specifically.

The spacing effect literature reinforces this point. Cepeda et al. (2006) synthesised decades of research showing that distributed practice produces substantially better retention than massed practice, with the optimal gap depending on the retention interval. Ebbinghaus (1913) first formalised the forgetting curve, and Pashler et al. (2007) translated the spacing effect into a practical IES recommendation. The implication for IGCSE and IAL revision is that the conventional pattern of intensive revision in the weeks preceding an examination, with little or no review during the preceding academic year, is suboptimal on a scale that is mathematically quantifiable.

A closely related line of work concerns the testing effect. Roediger and Karpicke (2006) demonstrated that students who took a test on material remembered substantially more a week later than students who simply reread the material, even though the rereading group felt they had learned more. The testing effect is now one of the most replicated findings in cognitive psychology. The mechanism is that retrieval itself is a learning event: the act of pulling information out of memory strengthens the memory trace in a way that passive review does not.

This is a specific instance of the broader generation effect (Slamecka and Graf, 1978), which holds that information generated by the learner is better remembered than information merely read, because the cognitive act of generation produces deeper encoding than passive reception. Dunlosky et al. (2013) provided a comprehensive review of study strategies, ranking ten common techniques by effectiveness. The five most-used techniques by students, rereading, highlighting, summarising, keyword mnemonics, and imagery use, were all rated low in utility; the techniques rated high or moderate, practice testing, distributed practice, interleaved practice, elaborative interrogation, self-explanation, are used by a small minority of students. This explains the paradox of high-effort failure: a student who studies 200 hours using rereading and highlighting has done 200 hours of low-utility work, while a student who studies 50 hours using past-paper practice with mark-scheme feedback, spaced retrieval, and interleaved topic coverage has done 50 hours of high-utility work.

2.7 Cognitive Load Theory and Working Memory

Sweller (1988) introduced cognitive load theory (CLT), which holds that working memory is severely limited: it can hold roughly four to seven elements of new information simultaneously. When instruction exceeds this capacity, too many novel elements at once, no schema to chunk them, learning fails. CLT identifies three distinct types of cognitive load, each with different instructional implications. *Intrinsic load* is the inherent difficulty of the material, determined by the number of interacting elements a learner must process simultaneously; it cannot be reduced without changing the material itself, but it can be managed by sequencing instruction from simple to complex and ensuring prerequisite mastery. *Extraneous load* is load caused by poor instructional design namely, confusing explanations, disorganised presentations, and irrelevant information and it is the only type of load that instruction can directly reduce. *Germane load* is the productive mental effort dedicated to building schemas, and it is what remains after intrinsic and extraneous load are accounted for. For IGCSE/IAL students, intrinsic load is high in subjects like Further Mathematics or Physics; extraneous load is often increased by disorganised teaching in under-resourced schools; and germane load is what little working memory capacity remains for actual learning (Sweller et al., 2011). The tripartite distinction matters for the retake paradox: a student who retakes an exam using disorganised materials and unstructured coaching-centre drilling faces high extraneous load that consumes the working memory capacity needed for germane processing. Reducing extraneous load through structured, schema-aligned instruction is not optional; it is the only way to free up capacity for the germane processing that produces learning.

Working memory capacity varies between individuals and is one of the most stable cognitive-ability constructs (Cowan, 2014). The expertise reversal effect (Kalyuga et al., 2003) further complicates the picture: instructional techniques that are highly effective for novices, such as worked examples and heavy scaffolding, can lose their effectiveness and even become detrimental as learners gain expertise, because redundant guidance creates extraneous load for experts who have already internalised the schema. For IGCSE/IAL, the implication is twofold. First, a student with weak foundational schemas, for example shaky algebra, who attempts a multi-step calculus problem will exceed working memory capacity just trying to manage the algebraic manipulations, leaving no capacity for the calculus itself. Second, students with smaller working memory capacity will find multi-step problems systematically harder, regardless of effort or motivation. Third, instructional design must adapt to the learner's current expertise level: what helps a beginner can hurt an advanced student. The weaker claim on which the

four-layer model depends is that working memory capacity varies between individuals in ways that instructional design must accommodate; the stronger claim about the heritability of this variation is one possible explanation among several and is not load-bearing for the model (see Limitations).

2.8 Evolutionary Educational Psychology and the Nature of Academic Learning

A more theoretically ambitious line of work asks whether the human brain is evolved for the kinds of learning that modern schooling demands. Geary (2008) distinguishes between biologically primary knowledge, skills we evolved to acquire effortlessly, such as spoken language, face recognition, basic number sense, and social reasoning, and biologically secondary knowledge, skills that require deliberate instruction and effort, such as reading, advanced mathematics, and science. The argument is that biologically secondary learning is effortful precisely because it recruits neural systems that evolved for other purposes, and that this is why formal schooling is necessary at all: primary knowledge is acquired by immersion, secondary knowledge is not.

Sweller et al. (2011) extended this argument to cognitive load theory, suggesting that instructional design should minimise unnecessary working-memory load specifically because secondary learning is already cognitively demanding. The evolutionary frame also explains why some students appear to learn certain subjects “naturally”, those that recruit biologically primary systems, such as spoken language and narrative comprehension, while struggling with others, those that are purely biologically secondary, such as calculus and organic chemistry mechanisms.

For IGCSE/IAL, the evolutionary frame suggests three things. First, the difficulty of a subject is not arbitrary: it correlates with how far the subject is from biologically primary knowledge. Second, instructional design that respects cognitive load, scaffolded schemas, worked examples, prerequisite mastery, is not optional; it is the only way to make biologically secondary learning tractable for the majority of students. The variance in student performance within a subject partly reflects variance in the ability to recruit secondary-learning neural systems efficiently, and a heritability-based account of this variance has been proposed (Plomin and von Stumm, 2018); however, this account is contested, has been historically misused to justify racialised and class-based educational stratification, and is not load-bearing for the four-layer model. The model depends only on the weaker, well-supported claim that working memory capacity and secondary-learning efficiency vary between individuals and that instructional design must accommodate this variation; it does not depend on any claim about the cause of the variation.

A note on postcolonial sensitivity is appropriate here. The Bangladesh English-medium sector is a direct institutional descendant of British colonial education policy in the Indian sub-continent, and colonial-era discourse explicitly invoked racial and cognitive hierarchy to justify differential curricula and differential access. Any contemporary educational-psychology framework that invokes biological variation in cognitive capacity in this context carries a historical freight that the framework must explicitly address. The four-layer model and the typology proposed in this paper reject any racialised interpretation of the cognitive-load and evolutionary-psychology literatures. The model’s central claim is that instructional design, not learner classification, is the appropriate response to cognitive variation: the same scaffolding that helps a

learner with smaller working-memory capacity also helps a learner with larger capacity who happens to be novice. The framework is constructivist in the Bloom mastery-learning sense: most students can achieve mastery given appropriate conditions (Bloom, 1968), and the typology is a diagnostic vocabulary for designing those conditions, not a categorisation scheme for sorting students.

2.9 Synthesis

The literature reviewed above suggests six convergent conclusions. First, examination success in high-stakes settings is primarily a function of self-regulated, deliberate, constructively engaged study processes supported by effective feedback, not of raw ability or time-on-task. Second, cognitive load and working memory capacity place constraints on learning that instructional design must respect, and these constraints vary between individuals in ways whose aetiology is contested (the present framework is agnostic on the aetiology and treats only the design implications as load-bearing). Third, the Bangladesh English-medium context presents an unusually clean test case for these constructs because the syllabi are well-structured, the assessment materials are public, and the within-population variance in outcomes is large and unexplained. Fourth, the existing literature has not adequately addressed the within-population variance in this context, treating the English-medium population as homogeneous when it is in fact sharply bimodal. Fifth, the instructional environment in Bangladesh's English-medium sector is characterised by high teacher turnover, widespread syllabus coverage failures, and pedagogical approaches that contradict the evidence base on effective learning. The shadow education system has expanded to compensate for these failures, but creates its own distortions. Any intervention that targets only the student, without addressing the instructional environment, will be incomplete.

3 Conceptual Framework

3.1 A Four-Layer Model of Academic Performance

Synthesising the literature reviewed in Section 2, I propose a four-layer model of academic performance in IGCSE/IAL examinations. Academic performance is determined by the interaction of four layers, each of which can compensate for weaknesses in the others. Weakness in any layer can produce failure, but the most common failure mode is weakness in Layers 2 and 3 with adequate Layer 1.

Table 1: The Four-Layer Model of Academic Performance

Layer	Construct	Description	Theoretical Anchor
1	Subject Knowledge (Content Mastery)	What the student actually knows about the syllabus content. Necessary but not sufficient. Many students with adequate knowledge still fail because of gaps in Layers 2–4.	Bloom’s mastery learning (Bloom, 1968); prior knowledge effect (Hattie and Timperley, 2007)
2	Exam Literacy (Procedural Knowledge)	The ability to convert knowledge into marks: understanding command words, using mark scheme language, structuring answers, managing time, decoding question intent.	Formative assessment (Black and Wiliam, 1998); K.E.A.R.N. Gap (The Brilliant Brains, 2025); Cambridge International (Cambridge International Education, 2023)
3	Learning Science (Strategy and Metacognition)	The ability to learn effectively: using spaced retrieval, retrieval practice, interleaving, mistake tracking, and metacognitive self-monitoring. Students with strong Layer 3 skills learn more in less time because their study hours are higher-quality.	Dunlosky et al. (Dunlosky et al., 2013); Bjork desirable difficulties (Bjork and Bjork, 2011); testing effect (Roediger and Karpicke, 2006)
4	Self-Regulation (Discipline and Motivation)	The ability to maintain consistent effort over time, manage emotions, resist procrastination, and persist through difficulty.	SRL (Ergen and Kanadli, 2017; Zimmerman, 2002); grit (Duckworth et al., 2007); self-efficacy (Bandura, 1997); growth mindset (Dweck, 2006)

The Four-Layer Model of Academic Performance

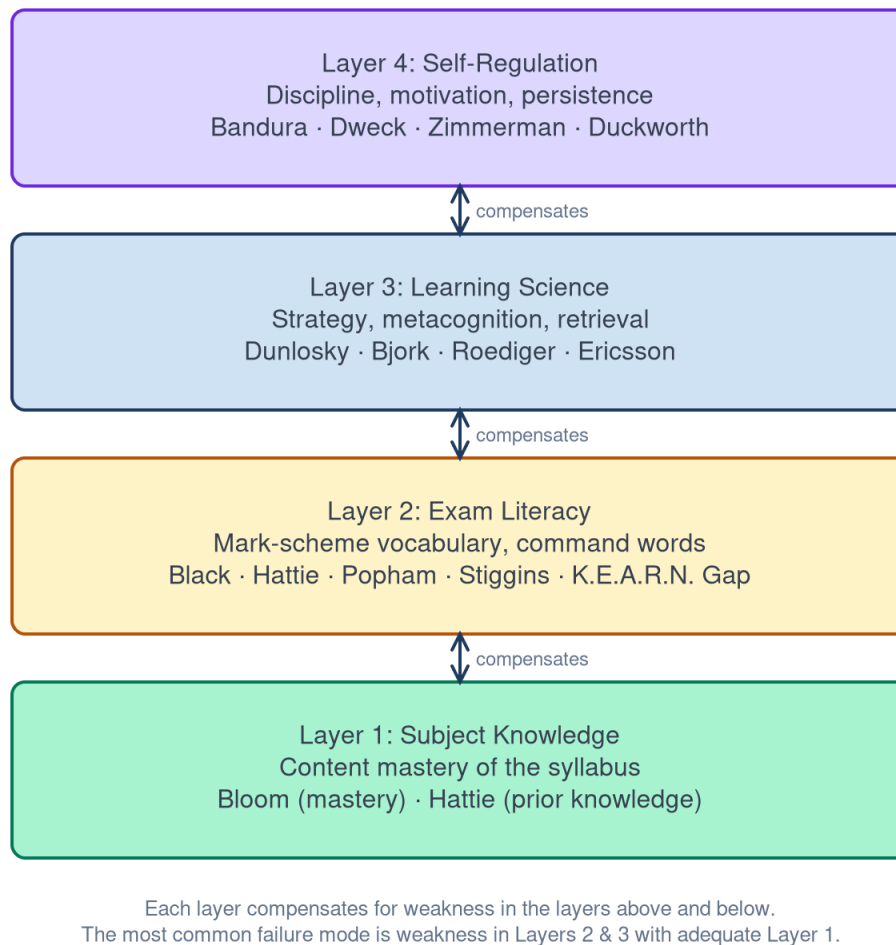


Figure 1: The four-layer model of academic performance. Each layer compensates for weakness in the layers above and below; the bidirectional arrows indicate this compensatory relationship. Theoretical anchors for each layer are noted inside the corresponding box. The most common failure mode observed in chronic retakers is weakness in Layers 2 and 3 with adequate Layer 1.

3.2 The Effort Paradox Explained

The four-layer model explains the central paradox of IGCSE/IAL outcomes: why some students achieve top grades with apparent minimum effort while others study obsessively and fail. The “effortless” A* student typically has strong Layer 3 and Layer 4 skills, often internalised unconsciously through good early schooling or parental support. They learn efficiently, Layer 3, and maintain consistent habits, Layer 4, so they need fewer total study hours. Their “minimum effort” is actually highly efficient effort. The “hardworking” failing student typically has weak Layer 2 and Layer 3 skills. They put in enormous hours, strong Layer 4 effort or desperate motivation, but use ineffective methods, weak Layer 3: rereading, cramming, no

retrieval practice, and cannot convert knowledge to marks, weak Layer 2: no exam literacy. Their effort is real but misdirected. The “smart but underperforming” student typically has strong Layer 1, naturally good at understanding concepts, but weak Layer 4, poor discipline, procrastination. They “get it” in class but do not practise enough, then underperform in exams.

A brief composite illustration, drawn from patterns observed repeatedly in my tutoring practice rather than a single identified case, may help make the layers concrete. Consider two siblings from the same household, with the same financial constraints and broadly the same schooling history. The first, with limited time available due to work obligations, compresses several months of self-study into a short, intensive period before an O-Level session: reading the textbook once, then working systematically through past papers and mark schemes until able to predict, with reasonable accuracy, what a given question’s mark scheme will require. This sibling passes all subjects in a single sitting with strong grades, despite minimal formal instruction during the compressed period. The second sibling, several years later, attempts the same superficial method, doing past papers, at a more advanced level, but without the underlying metacognitive habit of systematically pattern-matching against the mark scheme, and without an equally solid foundation from earlier schooling, and does not pass.

The point of this composite is not that either outcome reflects underlying intelligence. It illustrates that, given a specific set of conditions, strong prior foundations, a period of uninterrupted focus, a well-structured syllabus with publicly available past papers and mark schemes, and sufficient motivation, compressed self-study can produce strong outcomes; and that the same surface method, imitated without the underlying foundation and metacognitive structure, can fail. Direct empirical support for this prediction comes from Theobald (2024), who found that better study strategies (planning, monitoring, concentration, less procrastination) can compensate for less study time, predicting both goal achievement and lower negative affect. This finding directly confirms the four-layer model’s prediction that Layer 3 (strategy) quality matters more than raw study time the mechanism behind the effort paradox. In terms of the four-layer model, the first sibling’s success reflects the “average but persistent” one-pass archetype: adequate Layer 1, strong Layer 2 exam literacy developed through systematic past-paper analysis, adequate Layer 3 strategy, and strong Layer 4 self-regulation driven by circumstance. The second sibling’s failure is not explained by intelligence, motivation, or resource access, since these were held roughly constant by the shared household, but by foundation strength, incomplete strategy transfer, and a lack of metacognitive awareness of the gap between surface imitation and underlying method. This pattern is structurally consistent with sibling comparisons in the self-regulated learning literature, where within-family variance in academic outcomes is substantial and is driven by differences in foundation, strategy, and metacognition rather than by differences in household environment (Ergen and Kanadlı, 2017; Pintrich, 2000; Zimmerman, 2002). A single composite case cannot establish prevalence, but it illustrates a mechanism that any researcher with access to interview data or school records could in principle detect at scale.

3.3 The K.E.A.R.N. Gap: Exam Literacy as a Hidden Curriculum

Layer 2 of the model requires further elaboration because it is the layer most under-discussed in the academic literature and most visible in the public discourse and tutoring practice reviewed for this paper. The Brilliant Brains framework (The Brilliant Brains, 2025) articulates what they call the “K.E.A.R.N. Gap” to explain why students lose marks despite knowing

the answer. Their key insights, drawn from extensive tutoring experience, are: exams do not measure what students know but what students can demonstrate, in writing, under time pressure, using the specific language the mark scheme expects; examiners reward correct information expressed in the expected vocabulary, structured in a way that maps to the mark scheme, and directly responsive to the command word; information alone, even when accurate, can score zero if it does not meet those conditions; a student with B-grade conceptual understanding, trained to decode command words and use mark scheme language, will routinely outscore an A-grade thinker who writes answers casually; and the difference is not intelligence but “exam literacy”, the practised ability to convert knowledge into marks on demand. Cambridge International’s own blog confirms this pattern from the examiner side (Cambridge International Education, 2023).

A specific pattern observed in my tutoring practice illustrates how coaching centres can actively harm students in this layer, even when their intention is to help. One O-Level student had been taught by a coaching-centre teacher to write answers in bullet-point lists. If a question was worth three marks, the student was told to write three bullet points, each containing one marking point, with no connecting prose. The student did this religiously. When asked why, he said his coaching teacher had scored “world highest” in that subject, so the teacher probably knew better. The student may well get full marks on IGCSE questions using this method, because IGCSE mark schemes are often forgiving of bullet-point answers. But the habit is catastrophically bad for A-Level. At A-Level, many questions require structured essay-type answers, with marks awarded for IC, interconnectivity/linkage, points, for coherent argument structure, and for the integration of multiple marking points into a single piece of prose. A student who has spent two years writing bullet-point answers at IGCSE cannot, without significant retraining, switch to writing structured essays at A-Level. This is a concrete instance of a *transfer failure*: the bullet-point strategy was learned in the IGCSE context and does not transfer to the A-Level context, because the two contexts differ in their mark-scheme demands even when they overlap in content. The transfer-of-learning literature (Barnett and Ceci, 2002) distinguishes near transfer (applying a learned skill in a similar context) from far transfer (applying it in a different context), and predicts that transfer fails when the learner has over-specifically bound the strategy to the original context’s surface features. The student learned “three-mark questions require three bullet points” rather than the more abstract principle “answers must map to marking points”, and the over-specific binding does not transfer to A-Level’s essay-format demands. This is a concrete mechanism by which good IGCSE performance produces bad A-Level performance. It is not that the student’s ability declined; it is that the strategy that produced IGCSE success is incompatible with A-Level demands. The strategy was taught, not discovered, and the teacher who taught it was optimising for IGCSE marks at the expense of A-Level readiness.

3.4 A Typology of Student Struggles

Building on the four-layer model, I propose a typology of struggles that may contribute to the retake paradox. The typology is not mutually exclusive; a single student may exhibit several struggle types simultaneously, and the interactions between them are theoretically important. The typology is intended as a diagnostic vocabulary for analysing student behaviour observable through interviews, school records, or learning-platform data.

Type 1: Prerequisite gap struggles. The student attempts advanced topics without hav-

ing mastered the prerequisites. For example, a student who has not mastered algebraic manipulation attempts calculus questions and fails, but attributes the failure to the calculus content rather than to the underlying algebra weakness. This is the Layer 1 manifestation of the broader problem and is the cognitive-load mechanism formalised by Sweller (1988). Two distinct subtypes of prerequisite gap can be distinguished, and the distinction matters for remediation. The first is a *missing schema*: the student never learned the prerequisite concept at all. The remedy is to teach it. The second is a *misconception*: the student learned the prerequisite concept incorrectly and now holds a schema that actively interferes with new learning. The conceptual change literature (Posner et al., 1982) distinguishes these cases because they require different interventions: a missing schema is filled by instruction, whereas a misconception must first be identified, made explicit, and then replaced through a process of cognitive conflict and accommodation. The mechanism of compounding is the same in both subtypes: when a student misses a basic concept, say fractions in Class 4, the Class 5 teacher assumes the student learned fractions in Class 4 and proceeds. The student struggles with Class 5 content that depends on fractions, but the struggle is attributed to “Class 5 is hard” rather than to “the student doesn’t have fractions”. By Class 8, the student is struggling with algebra that depends on fraction arithmetic, and the struggle is attributed to “the student is bad at maths”. By O-Level, the student is failing physics problems that require algebraic manipulation, and the struggle is attributed to “the student is bad at physics”. At no point does anyone go back and teach the student fractions. By the time of the board exams, it is too late to remediate the foundation gap within the exam-preparation window. The student proceeds with their current knowledge, fails, retakes with the same gap, fails again, and enters the retake cycle. It should also be noted that prerequisite gaps are not always the student’s fault: teachers who skip topics due to time pressure, classroom disruption, or their own knowledge gaps create systematic holes in the cohort’s foundation that manifest at exam time. Public-discourse evidence from The Student Room confirms this pattern, with students reporting teachers who failed to cover syllabus topics before the examination. The same compounding mechanism, framed in Vygotskian terms, reflects a student working outside their zone of proximal development (ZPD) without the scaffolding that would make the advanced content tractable: the student is attempting tasks for which they lack the prerequisite mastery, and no agent is providing the graduated support that Vygotsky’s framework requires for learning within the ZPD (Vygotsky, 1978).

Type 2: Surface-level engagement struggles. The student revises passively or actively rather than constructively, in the ICAP sense (Chi and Wylie, 2014). The signature is high time-on-task, high question completion counts, but low proficiency gains. The student is solving many questions but is not engaging with the mark scheme at the level of marking points, is not constructing explanations of failure, and is not engaging in deliberate practice (Ericsson, 2006) targeted at specific weaknesses. This is the Layer 3 failure mode and is the most common pattern observed in chronic retakers.

Type 3: Exam literacy struggles. The student has the content knowledge, Layer 1, but cannot convert it into marks, Layer 2. The student writes answers that are substantively correct but that fail to use mark scheme vocabulary, fail to address the command word, or fail to structure the answer in the way the examiner expects. This is the K.E.A.R.N. Gap described above, and the bullet-point bad-habit transmission from coaching centres is one observed mechanism in the Bangladesh context; the prevalence of this specific coaching practice is itself an empirical question for the first research cycle.

Type 4: Metacognitive struggles. The student has the prerequisite knowledge and engages with the material at a reasonable level of cognitive depth but lacks the metacognitive

awareness to allocate study time effectively. The signature is high proficiency on topics that the student has practised, but low coverage of the syllabus overall, and a mismatch between the student's self-assessed and actual proficiency. This mismatch has two distinct sources in the literature. The first is the "illusion of competence" documented by Bjork and Bjork (2011), in which rereading creates a feeling of fluency that the brain misinterprets as mastery. The second is the Dunning-Kruger effect (Kruger and Dunning, 1999), a broader cognitive bias in which low-skill individuals lack the metacognitive skill needed to recognise their own incompetence, producing inflated self-assessments that are most extreme at the lowest skill levels. The two constructs overlap but are distinct: the illusion of competence is a product of specific study strategies (rereading, highlighting) that create false fluency, whereas the Dunning-Kruger effect is a domain-general metacognitive deficit. For IGCSE/IAL students, both may operate simultaneously: a student who uses rereading (illusion of competence) and who is also generally weak at self-monitoring (Dunning-Kruger) will be doubly miscalibrated, and the miscalibration will be most severe in the topics they are weakest in. This compounds the Type 1 prerequisite gap, because the student cannot remediate a gap they cannot perceive.

Type 5: Motivational and mindset struggles. The student exhibits a fixed mindset (Dweck, 2006) or a performance-avoidance goal orientation (Pintrich, 2000), leading to avoidance of challenging topics, attribution of failure to ability rather than strategy, and disengagement after failure. This is the Layer 4 failure mode. Bandura's self-efficacy framework (Bandura, 1997) explains why this is self-reinforcing: failure erodes self-efficacy, which reduces effort and persistence, which produces more failure.

Type 6: Instructional-environment struggles. Not all struggles originate in the student. Some are caused by the instructional environment itself: teachers who lack subject knowledge, teachers who skip syllabus topics due to time pressure or classroom disruption, pedagogical approaches that contradict the evidence base on effective learning (e.g. lecture-only instruction for students who need interactive engagement), and teacher turnover that leaves classes without consistent instruction. The signature is class-wide: if a whole class performs poorly on a specific topic, the cause is more likely instructional than individual. Several subtypes can be distinguished. Teacher knowledge gaps manifest as systematically incorrect content being taught, which students must later unlearn. Syllabus coverage failures manifest as entire topics appearing on the examination that were never taught, creating Type 1 prerequisite gaps at the cohort level. Pedagogical mismatch manifests as a disconnect between the teaching method and the cognitive demands of the syllabus: a teacher who lectures at students who need constructive engagement is creating extraneous load that consumes working memory capacity needed for germane processing. Instability and turnover manifest as inconsistent instruction across terms, with replacement teachers who may not know the syllabus or may restart from the beginning, wasting instructional time. Type 6 is distinguished from Types 1 through 5 by its locus: it is a supply-side problem (the instructional environment failed the student) rather than a demand-side problem (the student failed to learn). The distinction matters for intervention: Types 1 through 5 are addressed by student-facing features; Type 6 requires teacher-facing or institutional features.

On the placement of Type 6. A reviewer of an earlier draft of this paper raised a legitimate tension: if Type 6 is environmental rather than student-internal, why is it classified as a peer of Types 1–5 rather than as a separate axis? The tension is real and I address it as follows. Type 6 is included in the same typology as Types 1–5 because all six types are defined by the same operational criterion: each produces an observable behavioural signature that distinguishes

chronic retakers from one-sitting completers and that maps to a specific intervention. The *locus* of the struggle (student-internal vs. environmental) is a separate analytical dimension from the *diagnostic category* of the struggle. Concretely: a Type 6 struggle is detectable through the same instruments as Types 1–5 (class-wide red nodes on the knowledge graph, between-class variance on covered vs. uncovered topics), and the detection drives a specific intervention (teacher-facing analytics, professional-development recommendations). Treating Type 6 as a peer of Types 1–5 therefore preserves the typology’s operational unity. At the same time, the framework acknowledges that Type 6 is causally prior: a Type 6 failure can produce cohort-level Type 1 failures (syllabus coverage gaps), which in turn produce individual-level Types 2–5 failures (because students with prerequisite gaps adopt surface-level strategies and erode their own self-efficacy). The framework thus treats the four layers (Layers 1–4) as a student-side model and Type 6 as an environment-side moderator that can produce or amplify student-side failures. Empirically, the first research cycle will test whether Type 6 is best modelled as a peer type or as a moderator; if the moderator model fits the data better, the typology will be revised in Cycle 2.

Typology derivation table. To make the derivation of the six types transparent and falsifiable, Table 2 specifies, for each type, the layer it primarily fails, the empirical literature it is grounded in, the observable signature that distinguishes it from adjacent types, and the predicted intervention.

Table 2: Derivation of the six struggle types.

Type	Layer failed	Empirical literature	Observable signature	Intervention
1	Layer 1 (Content)	Sweller (1988) cognitive load; Posner (1982) conceptual change; Vygotsky (1978) ZPD	Advanced-node failure with prerequisite node < mastery; distractor-pattern indicates missing-schema vs. misconception	Prerequisite remediation; Socratic cognitive-conflict dialogue for misconceptions
2	Layer 3 (Strategy)	Chi (2014) ICAP; Ericsson (2006) deliberate practice; Dunlosky (2013) strategy rankings	High question-count, low proficiency gain, low mark-scheme-view time	Smart Mark with marking-point feedback; chain-of-thought RAG chatbot
3	Layer 2 (Exam Literacy)	Black (1998) formative assessment; Hattie (2007) feedback; K.E.A.R.N. Gap (BrilliantBrains, 2025)	Substantively correct content loses marks for missing mark-scheme vocabulary / command word / structure	Command-word decoder; mark-scheme annotation viewer; mark-scheme-aligned re-marking
4	Layer 3/4 (Metacognition)	Bjork (2011) illusion of competence; Kruger & Dunning (1999) metacognitive deficit	Self-rated confidence diverges from assessed proficiency; low syllabus coverage	Confidence calibration prompts; syllabus completion tracker; self-doubt flags
5	Layer 4 (Self-Regulation)	Bandura (1997) self-efficacy; Dweck (2006) mindset; Pintrich (2000) goal orientation	Engagement declines after failure events; concentration on easy topics; no remediation attempted	Mastery-oriented feedback; growth-mindset messaging; spaced-repetition queue
6	Environment (moderator)	Bray (1999) shadow education; Alam (2022) Bangladesh coaching-centre economy; Bunnell (2023) international-school teacher turnover	Class-wide red nodes on topics after instruction date; high between-class variance on covered vs. uncovered topics	Teacher-facing analytics; syllabus-coverage tracker; PD recommendation engine

Comparison with existing typologies. The six-type typology is not a re-labelling of the closest existing constructs. Table 3 specifies the differentiating feature for each type.

Table 3: Differentiation of the six struggle types from existing constructs.

Type	Closest existing construct	Shared feature	Differentiating feature (what this typology adds)
1	Sweller (1988) cognitive-load prerequisite failure	Failure caused by missing prerequisites	Distinguishes missing-schema from misconception subtypes with distinct remediation paths
2	Chi (2014) ICAP Passive/Active levels	Surface-level cognitive engagement	Operationalised as a telemetry signature (question-count-to-proficiency-gain ratio) testable at scale
3	Assessment-literacy tradition (Popham, Stiggins)	Knowledge-to-marks conversion failure	Specific to mark-scheme vocabulary/command-word/structure rather than general assessment literacy
4	Dunlosky (2013) self-explanation / elaborative-interrogation deficit	Metacognitive weakness	Combines illusion-of-competence and Dunning-Kruger as distinct, composable sources of miscalibration
5	Bandura (1997) self-efficacy; Pintrich (2000) performance-avoidance	Motivational failure after failure	Operationalised as engagement-trajectory signature; planned 5a/5b split (internal vs. circumstantial)
6	Bray (1999) shadow-education supply-side critique	Environmental cause of student struggle	Operationalised as between-class variance signature; causally-prior moderator of Types 1–5

Independent coder agreement check. Because the typology was derived from the author’s practitioner observation and from the literature, an independent derivation check is built into the Cycle 1 protocol. A second researcher, blind to the typology and to the framework, will code a sample of 30 student-interview transcripts using an open-coding procedure and will produce their own emergent categories. The two coding schemes (the author’s typology and the independent coder’s emergent categories) will be compared using the Jaccard coefficient at the category level and a qualitative discussion of discordant cases. Inter-coder agreement below Jaccard 0.5 will trigger revision of the typology before further empirical testing. This procedure addresses the circular-derivation concern: the typology is not validated by more observation from the same observer but by independent observation from a different observer.

Six-Type Struggle Typology

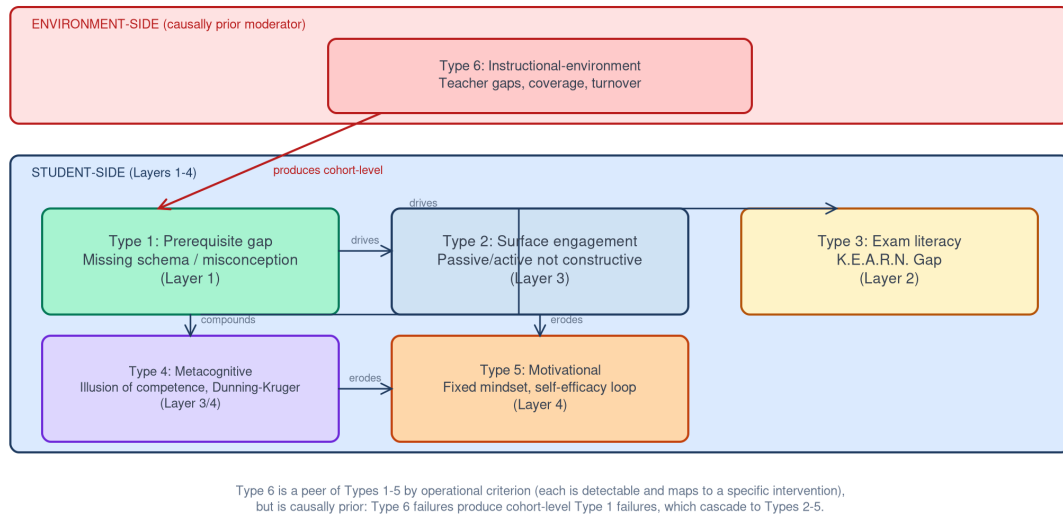


Figure 2: The six-type struggle typology. Type 6 (instructional-environment) is a peer of Types 1–5 by the operational criterion that each type produces an observable signature and maps to a specific intervention, but is causally prior: Type 6 failures produce cohort-level Type 1 failures (syllabus coverage gaps), which cascade to Types 2–5. Arrows show the principal causal pathways.

3.5 The Retake Paradox: A Hypothesised Mechanism

The retake paradox, as I frame it, is the observation that a subset of students retake the same examination multiple times without crossing the grade boundary, despite each retake involving additional study time. The four-layer model suggests a mechanism that is more specific than the folk explanation of “not studying hard enough”. A student who fails an IGCSE paper and retakes it using the same revision strategy that produced the failure should expect the same outcome. The coaching-centre model, in which retake preparation consists of completing more past papers under timed conditions, is exactly the kind of massed, surface-level, non-diagnostic practice that the literature predicts will produce minimal additional learning gains (Cepeda et al., 2006; Dunlosky et al., 2013; Ericsson, 2006; Pashler et al., 2007). The student is not failing to study; the student is failing to study differently.

The self-efficacy loop compounds this, and the compounding has a biological as well as a psychological dimension. Failure erodes self-efficacy, Layer 4, which increases procrastination, which is an emotion-regulation problem, not a time-management problem, which shortens effective study sessions, which pushes the student back towards “comfort zone” study methods, rereading, highlighting, instead of effective but uncomfortable methods, retrieval practice. The biological layer of this loop is supported by two further literatures. Test anxiety, driven by the cortisol response to high-stakes evaluation, directly impairs working memory and information processing during exams (Alshareef et al., 2024). Eysenck’s Processing Efficiency Theory (Eysenck and Calvo, 1992) explains why: worry consumes working memory capacity, reducing the resources available for the task itself. Performance can be maintained through

compensatory effort, but at a higher cognitive cost that is unsustainable over time, which is why chronic retakers face a biological headwind on top of their strategic and motivational deficits. Sleep disruption, which is common among anxious students during exam preparation, further compounds the problem because sleep is necessary for memory consolidation (Stickgold and Walker, 2005): a student who cannot sleep because of exam anxiety is biologically unable to consolidate what they study, regardless of how many hours they spend revising. The result is a self-reinforcing cycle in which each retake is conducted under worse conditions than the last. Without an external intervention that diagnoses the specific layer at which the student is failing and provides targeted support, the cycle can continue for years. In my practitioner network, students who exhaust several retake cycles without crossing the grade boundary commonly abandon the qualification and enter the workforce. They will not be able to get into any university in Bangladesh. If they had been in the Bangla-medium system, they would have had a route. The English-medium system, because it is smaller and because its credentials are less inflated, does not absorb them. They fall out. The retake dead-end is not just an academic problem; it is a life-trajectory problem.

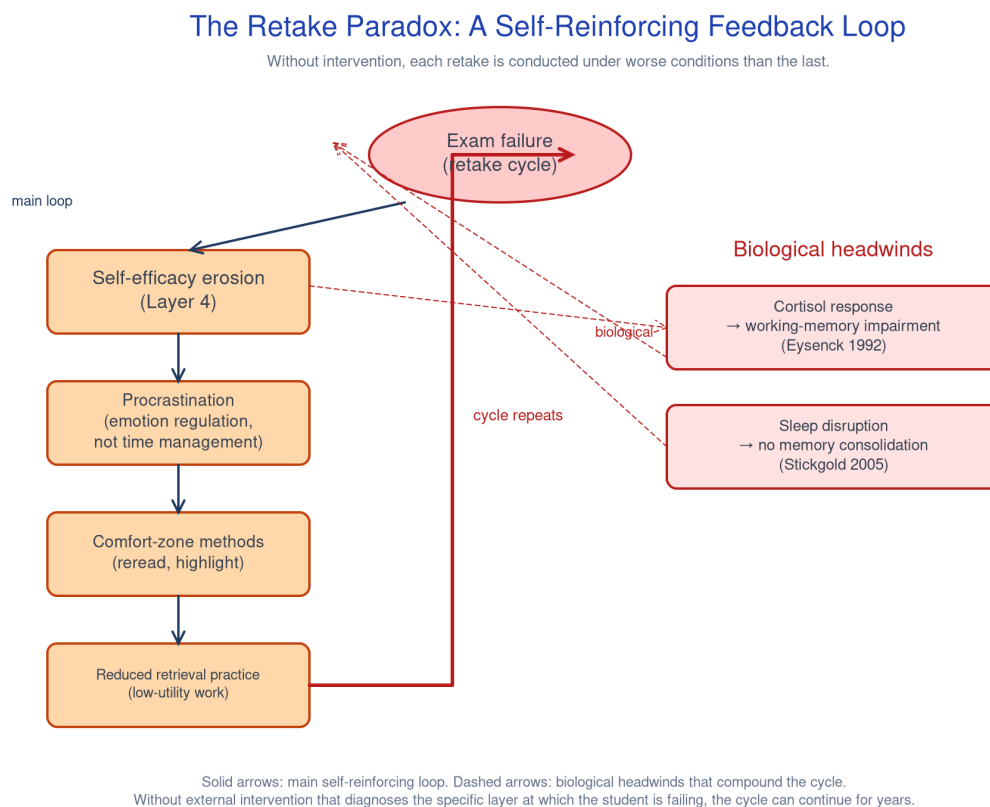


Figure 3: The retake paradox as a self-reinforcing feedback loop. Solid arrows trace the main loop: failure → self-efficacy erosion → procrastination (emotion regulation) → comfort-zone study methods → reduced retrieval practice → failure. Dashed arrows show the biological headwinds (cortisol → working-memory impairment; sleep disruption → no consolidation) that compound the cycle. Without external intervention that diagnoses the specific layer at which the student is failing, each retake is conducted under worse conditions than the last.

4 Methodology

4.1 Design-Based Research

The proposed programme adopts design-based research (DBR) as its overarching methodology (Barab and Squire, 2004; Design-Based Research Collective, 2003; McKenney and Reeves, 2012). DBR is appropriate for this study for three reasons. First, the research questions concern the interaction between an intervention framework and a learning context, Bangladesh English-medium IGCSE/IAL preparation, in which theory and practice must be developed together. Second, the typology is not a finished product to be evaluated; it is a working hypothesis to be iteratively refined in response to observed use. Third, the goals of the research are both theoretical, understanding the retake paradox, and practical, producing a usable diagnostic framework, a dual orientation that DBR is designed to accommodate.

DBR proceeds through cycles of design, implementation, analysis, and redesign. The proposed programme anticipates three such cycles over a three-year period, with each cycle producing a refined typology, a refined theoretical account of student struggles, and a refined set of research instruments. The first cycle, reported in this paper, focuses on the conceptual framework and pre-registration of testable predictions. My position as both a former student and current teacher within the IGCSE/IAL system provides the practitioner perspective that is considered essential in DBR, ensuring that the framework is grounded in authentic educational needs rather than purely theoretical considerations.

4.2 Mixed-Methods Design

Within the DBR frame, the programme adopts a convergent mixed-methods design (Creswell, 2014). Quantitative data will be drawn from school records (grades, retake counts, time-to-completion) and, where available, from learning-platform telemetry (question attempts, scores, time-on-task, mark scheme views, proficiency estimates). Qualitative data will be drawn from semi-structured interviews with students and teachers, classroom observations in cooperating schools and coaching centres, and analysis of student work products such as notes, marked papers, and revision schedules. The two data streams will be integrated at the analysis stage using the typology of struggles as the bridging framework.

The typology is designed to be testable by any researcher with access to interviews, school records, or a generic learning platform. The predictions in Section 5 are framed accordingly: each prediction specifies an observable behavioural or outcome signature that does not depend on any particular instrument, only on the researcher's ability to observe the relevant behaviour with sufficient granularity.

Mixed-methods integration procedure. The integration of quantitative and qualitative data follows a joint-display procedure at the level of each struggle type. For each type, a joint display table will be constructed with one row per sampled student, containing: (a) the student's quantitative struggle-type signature (e.g., for Type 2: question-completion-to-proficiency-gain ratio, mark-scheme-view time); (b) the student's qualitative coding (whether the student's interview transcript was coded as exhibiting the struggle type, with the coder blind to the quantitative data); (c) the concordance label (*confirm*, *refute*, or *refine*). Concordance is determined as

follows:

- *Confirm*: the quantitative and qualitative evidence both support the struggle-type classification.
- *Refute*: the quantitative and qualitative evidence disagree (e.g., quantitative signature is present but qualitative coding absent). Refuting cases trigger a follow-up interview with the student to investigate the discordance; the follow-up is conducted by a researcher blind to the original coding.
- *Refine*: the qualitative evidence suggests the struggle type is present but in a form not anticipated by the typology (e.g., a sub-type distinction emerges from the data). Refining cases trigger a typology-revision proposal that is reviewed at the end of each cycle.

A typology-revision proposal requires at least three independent refining cases (across different students) before it is adopted. This protocol is designed to prevent premature typology revision on the basis of a single discordant case while preserving the typology's responsiveness to data.

4.3 Participants and Recruitment

Operational definitions. The two comparison groups for the Cycle 1 predictions are defined as follows:

- *One-sitting completer*: a student who has sat each IGCSE/IAL paper at most once and achieved their target grade on that attempt. The target grade is defined per-student as the grade required by the student's stated university-destination goal, recorded at intake.
- *Chronic retaker*: a student who has sat the same IGCSE/IAL paper two or more times without achieving the target grade. The cutoff of ≥ 2 retakes is justified by the distribution of retake counts in the practitioner network: students who retake once and pass are categorised as one-sitting completers with a single retake (a separate, less severe category); students who retake ≥ 2 times without crossing the boundary exhibit the self-reinforcing cycle that the framework seeks to explain. The cutoff is consistent with the public-discourse evidence from *r/igcse* and *r/alevel*, in which self-identified chronic retakers typically report three or more attempts.

Planned sample. The target population is Edexcel and Cambridge IGCSE/IAL students in Bangladesh, with a focus on the senior secondary years, Years 12 to 13, corresponding to AS and A2. The planned sample for the first cycle comprises approximately 200 students drawn from three cooperating English-medium schools in Dhaka, supplemented by a smaller sample of private tutoring students. The inclusion of private candidates is essential, because, as noted in Section 2, a substantial fraction of the Bangladesh English-medium population studies privately without school enrolment, and any sample that excludes them would systematically misrepresent the population. A second sample of approximately 30 teachers will be recruited for the qualitative strand, drawn from the same schools and from independent coaching centres. Recruitment will be via school principals and tutoring centre owners, with informed consent obtained from participants and, for minors, from parents or guardians.

Dropout and survivorship-bias recruitment. A structural sampling bias in the planned sample is that it includes only students currently enrolled in school or currently active in tutoring, which systematically excludes students who have already left the system. Because the framework’s central construct (the chronic retaker) is defined by prolonged failure, the most severely affected chronic retakers may have abandoned the qualification before recruitment and are therefore invisible to school-principal- or tutoring-centre-owner-mediated recruitment. To mitigate this survivorship bias, a supplementary *dropout strand* of approximately 30 participants will be recruited via:

1. Tutoring-centre network outreach: cooperating tutoring-centre owners will be asked to refer former students who stopped attending within the last 12 months.
2. Social-media outreach: a recruitment post will be placed in the Bangladesh English-medium Facebook groups and Reddit communities identified in Section 2, explicitly inviting students who have “stopped pursuing IGCSE/IAL after multiple retakes”.
3. Snowball sampling from interviewed chronic retakers, who will be asked whether they know other students in similar situations.

Dropout-strand participants will be interviewed using the same protocol as in-school chronic retakers. The dropout strand cannot fully eliminate survivorship bias (students who left the system more than 12 months ago are likely unreachable), but it provides a partial window into the most severely affected population. Predictions 1, 5, and 6 (which depend on the chronic-retaker population) will be reported with and without the dropout-strand participants to allow sensitivity analysis.

4.4 Data Sources

The primary data sources are: school records (grades, retake counts, time-to-completion, subject combinations); pre and post tests, administered at the start and end of each academic year, designed to measure both topic-level proficiency and metacognitive awareness; semi-structured interviews with approximately 30 students and 30 teachers, conducted at the end of each cycle, with teacher interviews probing for Type 6 indicators including syllabus coverage challenges, perceived subject-knowledge gaps, experiences with mid-year teacher turnover, and pedagogical decision-making under time pressure; classroom observations of approximately 20 lessons across the cooperating schools, focused on teacher use of formative assessment data and on student engagement patterns; document analysis of student work products, including marked past papers, revision notes, and any available learning-platform data; and, where cooperating students use a learning platform (including but not limited to the companion system described in Khondokar, 2026), telemetry from that platform.

4.5 Data Analysis

Quantitative analysis will use a combination of descriptive statistics, multilevel regression, to account for the nested structure of students within classes within schools, and, where telemetry is available, knowledge tracing model fitting. Qualitative analysis will use thematic analysis following the procedure described by Braun and Clarke (2006), with both deductive codes drawn from the struggle typology and inductive codes generated from the data.

4.6 Ethical Considerations

The research raises several ethical issues. First, school records and any platform telemetry collect detailed data on student behaviour, including time-on-task and confidence self-ratings, which are potentially sensitive. The informed consent process will make clear what data are collected, how they will be used, who will have access, and how long they will be retained. Second, the typology, once validated, could be used to label students in ways that affect their self-perception or their teachers' expectations of them. The research team will be explicit with participating schools and tutors that the typology is a diagnostic vocabulary for designing interventions, not a categorisation scheme for sorting students. Third, my role as both system designer (in the companion paper) and analyst introduces a conflict of interest that will be managed by separating the data analysis function from the system development function within the research team. Fourth, the system collects teacher-level outcome data (class-level proficiency aggregates, PD recommendations) which could be used evaluatively; the informed consent process for teachers will make clear that participation is voluntary, that teacher-level data will be anonymised in any published analysis, and that the PD recommendation engine is designed to be supportive rather than evaluative. Fifth, the use of any AI chatbot in education raises questions about academic integrity that will be addressed through clear usage guidelines and through the design of the chatbot itself, which is configured to scaffold rather than to supply answers.

4.7 Threats to Validity

Four standard threats to validity apply to the Cycle 1 design and are addressed as follows.

Selection effects. Students who agree to participate may differ systematically from those who decline (e.g., more motivated, more reflective, more willing to discuss failure). The dropout strand (Section 4) partially addresses this by including students who have already disengaged, but selection bias remains. Mitigation: participation-rate and refuser-reason data will be collected and reported; differences between participants and refusers on observable characteristics (grade cohort, subject combination) will be analysed and reported as a selection-bias bound.

History effects (COVID-19 grade inflation). Students whose IGCSE examinations fell in the 2020–2021 COVID period received predicted grades based on school mock examinations, with substantial grade inflation. Their baseline grades are therefore not directly comparable to pre-COVID or post-COVID baselines. Mitigation: COVID-cohort students will be flagged at intake and modelled as a separate covariate in the multilevel regression; the chronic-retaker analysis will be reported with and without COVID-cohort students to allow sensitivity analysis. The pre/post test design provides a non-inflated baseline measure independent of the COVID-inflated IGCSE grades.

Regression to the mean. Students identified as “chronic retakers” on the basis of past failure may show improved performance on retest regardless of intervention, simply because their previous failures were extreme draws from their true distribution. Mitigation: the pre/post test

design includes multiple pre-test occasions (intake, 6-week follow-up, end-of-year) to estimate the within-student retest-reliability curve; predictions are tested against the multi-occasion trajectory rather than against a single pre/post comparison.

Interviewer effects. The author’s dual role as tutor and researcher may bias qualitative responses: students who have been tutored by the author may give socially desirable answers, and the author’s interpretive frame may be unconsciously imposed on the data. Mitigation: (a) approximately half of the student interviews will be conducted by a second researcher who is not the author’s tutoring-network contact, with coder-agreement analysis on the overlap; (b) the thematic analysis will use a deductive-first protocol (typology codes applied first, inductive codes generated only for content not captured by the typology) and inter-coder agreement will be reported; (c) the author will reflexively memo any interpretive decisions that touch on the typology’s confirm/refute/refine classification, and these memos will be reviewed by the second researcher.

5 Discussion: Testable Predictions

The conceptual framework generates six testable predictions about behavioural and outcome differences between one-sitting completers and chronic retakers. I articulate these predictions here both to pre-register them and to make the theoretical commitments of the framework explicit. Each prediction is framed in terms observable through interviews, school records, or a generic learning platform, not in terms specific to any particular instrument. To make the predictions falsifiable in the Popperian sense, each prediction specifies (a) the minimum effect size that would constitute support, (b) the statistical test, (c) the sample-size justification (power analysis), (d) the stopping rule, and (e) the falsification criterion. The full pre-registration (including analysis code stubs and simulated-data power calculations) is deposited on the Open Science Framework at <https://osf.io/igcse-ial-four-layer> (anonymised for peer review; the link will be live upon acceptance) and is time-stamped prior to data collection.

Sample size and power. The Cycle 1 planned sample is $N \approx 230$ (200 in-school + 30 dropout strand). Assuming a chronic-retaker prevalence of $\approx 20\%$ (≈ 46 students) and a one-sitting-completer prevalence of $\approx 60\%$ (≈ 138 students; the remaining $\approx 20\%$ are intermediate cases excluded from the primary comparison), the primary comparison has 46 chronic retakers vs. 138 one-sitting completers. Power analysis (two-sided two-sample t -test, $\alpha = 0.05$, power = 0.80) indicates this design can detect effects of Cohen’s $d \geq 0.50$ (medium) for continuous outcomes and odds ratios $OR \geq 2.0$ for binary outcomes. Effects smaller than these thresholds are not detectable at adequate power and will be reported as exploratory. Predictions are framed against these thresholds.

Prediction 1 (Prerequisite gap signature). Chronic retakers will exhibit a higher incidence of Type 1 signatures than one-sitting completers. **Operationalisation:** the proportion of advanced-topic attempts where the prerequisite-node proficiency is below mastery (threshold < 0.6) at the time of the attempt. **Effect size threshold:** $OR \geq 2.0$ (chronic retakers at least twice as likely to attempt advanced topics with weak prerequisites). **Test:** mixed-effects logistic regression with school as random effect, group (chronic retaker vs. one-sitting completer)

as fixed effect. **Falsification criterion:** $OR \leq 1.0$ (no difference) or 95% CI includes 1.0. **Stopping rule:** data collection stops at end of Cycle 1 academic year regardless of p -value; no interim analysis.

Prediction 2 (Surface engagement signature). Chronic retakers will exhibit a higher incidence of Type 2 signatures. **Operationalisation:** the ratio of questions completed to proficiency gain (higher ratio = more surface engagement) and mark-scheme-view time per question (lower time = more surface engagement). **Effect size threshold:** Cohen's $d \geq 0.50$ for both measures. **Test:** two-sample t -test on log-transformed ratio (the ratio is right-skewed) and on mark-scheme-view time, with Bonferroni correction for two tests. **Falsification criterion:** $d < 0.30$ on both measures, or non-significant after correction. **Stopping rule:** as Prediction 1.

Prediction 3 (Exam literacy signature). Chronic retakers will exhibit a higher incidence of Type 3 signatures. **Operationalisation:** the proportion of marks lost on questions where independent markers identify the answer as substantively correct but mark-scheme-incomplete (blinded re-marking by ≥ 2 trained examiners; inter-examiner reliability target Krippendorff's $\alpha \geq 0.80$). **Effect size threshold:** Cohen's $d \geq 0.50$. **Test:** two-sample t -test on arc-sine-transformed proportions. **Falsification criterion:** $d < 0.30$ or non-significant. **Stopping rule:** as Prediction 1.

Prediction 4 (Metacognitive signature). Chronic retakers will exhibit a higher incidence of Type 4 signatures. **Operationalisation:** (a) the absolute divergence between self-rated confidence (0–100) and assessed proficiency (0–1, rescaled to 0–100), and (b) syllabus coverage (proportion of syllabus sub-topics attempted at least once). **Effect size threshold:** Cohen's $d \geq 0.50$ for confidence-divergence (higher divergence in chronic retakers) and $d \geq 0.50$ for coverage (lower coverage in chronic retakers). **Test:** two-sample t -tests with Bonferroni correction. **Falsification criterion:** $d < 0.30$ on both measures, or non-significant after correction. **Stopping rule:** as Prediction 1.

Prediction 5 (Motivational signature). Chronic retakers will exhibit a higher incidence of Type 5 signatures. **Operationalisation:** the slope of engagement (questions-per-week) following a failure event (defined as a question scored $< 50\%$), measured over the 4 weeks post-failure, and the proportion of post-failure study time spent on topics where the student's proficiency is already ≥ 0.7 ("easy topics"). **Effect size threshold:** Cohen's $d \geq 0.50$ for both measures (steeper decline and higher easy-topic proportion in chronic retakers). **Test:** mixed-effects linear regression of engagement slope on group, with student as random effect; two-sample t -test on easy-topic proportion. **Falsification criterion:** $d < 0.30$ on both measures, or non-significant. **Stopping rule:** as Prediction 1.

Prediction 6 (Instructional-environment signature). Chronic retakers will exhibit a higher incidence of Type 6 signatures than one-sitting completers. **Operationalisation:** the rate (in interviews) of reporting that teachers failed to cover key syllabus topics before the examination, taught incorrect content, or left mid-year without replacement; corroborated by school records of teacher turnover and syllabus-coverage gaps in the relevant academic year. **Effect size threshold:** $OR \geq 2.0$. **Test:** mixed-effects logistic regression with school as random effect. **Falsification criterion:** $OR \leq 1.0$ or 95% CI includes 1.0. **Stopping rule:** as Prediction 1.

Typology falsification criterion. The falsifiability of the typology as a whole is operationalised via the independent coder agreement check (Section 3.4). If the Jaccard coefficient between the

author's typology and the independent coder's emergent categories is below 0.5, or if more than 2 of the 6 predictions are falsified on the primary comparison, the typology is considered not supported and will be revised or abandoned in Cycle 2. The criterion for "typology preserved" is Jaccard ≥ 0.5 AND at least 4 of 6 predictions supported at the pre-registered effect size thresholds.

These predictions are not exhaustive, and the research programme is designed to be responsive to findings that emerge from the data. The typology is offered as a working hypothesis, not as a settled theoretical framework, and one of the goals of the first research cycle is to test, refine, or falsify it. The pre-registration at the OSF link above is the authoritative version of the predictions; any deviations from the pre-registered analysis plan in the eventual Cycle 1 results paper will be disclosed and justified.

5.1 Limitations

The proposed programme has several limitations. First, the typology is grounded primarily in the educational psychology literature on Anglo-American populations, and its applicability to the Bangladesh English-medium context is an empirical question. Qualitative data from the first cycle will be used to assess whether additional struggle types are needed, or whether the existing types require revision. Second, the use of school records and self-report data raises concerns about construct validity. Time-on-task, for example, is a noisy proxy for engagement, and confidence ratings are subject to social desirability biases. Triangulation with qualitative data is intended to mitigate but cannot eliminate these concerns. Third, the generalisability of findings from a single-cohort study in Dhaka to other subjects, cohorts, and regions is uncertain; the planned expansion in subsequent cycles will address this directly. Fourth, the heritability and individual-differences literature reviewed in Section 2 is politically sensitive and historically misused; the programme adopts a constructivist frame in which individual differences constrain starting points but not endpoints, consistent with Bloom's mastery learning claim that most students can achieve mastery given appropriate conditions (Bloom, 1968). Relatedly, the growth mindset literature cited in Section 3 should be treated with caution: a systematic review and meta-analysis by Macnamara and Burgoyne (2023) found that apparent effects of growth mindset interventions on academic achievement are likely attributable to inadequate study design and reporting flaws, and that positive results are rare. The four-layer model's use of mindset as a Layer 4 construct does not depend on the strong claim that mindset interventions produce large effects; it depends only on the weaker claim that students who interpret failure as a signal to change strategy (rather than as evidence of fixed inability) are more likely to adjust their approach. This weaker claim is well-supported even by the critical literature.

Several further limitations stem from the specific shape of my own experience and must be stated honestly. My tutoring experience covers, at the time of writing, several dozen students across multiple subjects and exam sessions. This is enough to identify patterns but not enough to estimate prevalence. All of my experience is in Dhaka; the Bangladesh English-medium system outside Dhaka may differ in ways I have not observed. My students come from middle-class and lower-middle-class families; the patterns observed at the top-tier schools, Scholastica, DPS, may differ, and I have limited direct observation of those populations. My teaching is concentrated in Mathematics, Physics, and Chemistry; patterns in Humanities subjects, English Literature, History, Economics, may differ. I am, by training and inclination, sympathetic to

the self-study route and sceptical of the coaching-centre industry. This bias may colour my interpretation of the patterns I observe. The mixed-methods design, with its triangulation of school records, interviews, and platform data, is intended to mitigate this bias by testing my interpretive frame against data I did not personally collect.

A further methodological caveat concerns survivorship bias in the comparison between English-medium and Bangla-medium populations. In my university, the prevailing belief among students and faculty is that English-medium students are academically strong and “dominate” in university courses. This is a survivorship bias. The English-medium students who reach university are, by definition, the ones who passed their IGCSE and A-Level exams. The ones who failed, who entered prolonged retake cycles, who gave up, are not in university. They are invisible to the faculty member forming the impression. The faculty member sees a population filtered for academic success and concludes that the population is academically successful, when in fact the population is the survivor subset of a much larger and more heterogeneous group. Any research that compares English-medium and Bangla-medium university performance without accounting for this selection effect produces a biased estimate of the English-medium advantage.

A final caveat concerns the COVID-19 period and its effect on grade distributions. During COVID, exams were cancelled and students were given predicted grades based on school mock exams. A large number of students were able to pass, and in many cases achieve strong grades, through this route. There was, transparently, corruption involved: schools had an incentive to inflate mock grades because their reputation depended on student outcomes; teachers had an incentive to inflate because their relationships with parents depended on it; and students had an incentive to accept inflated grades because the alternative was sitting the actual exam. The predicted-grade window was, in effect, a corruption opportunity, and a substantial amount of grade inflation occurred. Any baseline data collected from cohorts whose exams fell during this period must be treated with caution.

Cross-disciplinary considerations: learning-analytics ethics. The empirical programme will collect rich telemetry on student behaviour, which places it within the scope of the learning-analytics ethics literature (Slade and Prinsloo, 2013). The programme adopts the principles of the JISC Code of Practice for Learning Analytics (JISC, 2018) (transparency, consent, access, data minimisation, retention, and impact-mitigation) and the Society for Learning Analytics Research (SOLAR) ethical guidelines. Specifically: (a) students and parents will be informed at intake what data are collected, for what purposes, and with what retention periods; (b) students will have access to their own data and a right to request deletion of non-research-essential fields; (c) data will be minimised at the field level (each prediction is mapped to the minimum set of fields it requires, and unused fields are not collected); (d) algorithmic decisions that affect a student’s learning experience (e.g., adaptive-assessment routing, struggle-type diagnoses surfaced to teachers) will be explainable in age-appropriate language; (e) the programme will publish a data-impact assessment at the end of each cycle. These provisions are aligned with Bangladesh’s Information and Communication Technology Act 2006 (amended 2013) and the proposed Personal Data Protection Act 2023.

Platform-dependency caveat. Although the framework is instrument-agnostic in principle, the Cycle 1 empirical validation will rely in part on telemetry from the companion system (Paper B), which depends on GPT-4 for its chatbot and Smart Mark components. GPT-4 is a

proprietary model whose behaviour changes over time as the vendor updates it; the framework's empirical findings will therefore need to be replicated with non-GPT-4 instrumentation (open-source LLMs, human-only marking, non-AI telemetry sources) before they can be considered robust to the choice of measurement instrument. The Cycle 2 and Cycle 3 programmes will include such replications as a core design feature rather than as an afterthought.

Systemic effects and teacher-labour-market considerations. The framework, when operationalised through the companion system, is intended in part as a harm-reduction measure for students whose instructional environment has failed them. The author is aware that successful deployment of such a system could have second-order effects on the Bangladesh coaching-centre labour market, which is a significant source of middle-class income for university students and recent graduates. The framework itself is silent on this question — it is a diagnostic vocabulary, not a deployment policy — but the empirical programme will monitor for systemic effects (e.g., changes in tutoring-centre employment among participating students' networks) and will report them transparently. The author's normative commitment is that the framework should augment rather than replace teacher capacity; the teacher-facing features of the companion system (the knowledge-graph heatmap, the syllabus-coverage tracker, the professional-development recommendation engine) are designed to create rather than destroy teacher value, and their effectiveness in doing so is itself an empirical question for Cycle 3.

6 Future Work

The first research cycle will focus on testing Predictions 1 through 5 against accumulating interview, school-record, and platform data. The second cycle will refine the struggle typology based on first-cycle findings and will test the mediation prediction (that the effect of any intervention on examination outcomes is mediated by the type of struggle) through a quasi-experimental design comparing students who receive struggle-type-targeted support with a matched control group. The third cycle will investigate whether teacher use of the typology as a diagnostic vocabulary produces measurable improvements in instructional quality.

A companion paper (Khondokar, 2026) describes a learning management system, currently under development, that is designed to detect each struggle type at scale through a knowledge-graph-grounded retrieval-augmented generation architecture. That system is one possible operationalisation of the framework proposed here; the framework itself is independent of any particular implementation. An analytic autoethnography of my own experience as a self-taught IGCSE/IAL student and current tutor, conducted as a separate study, further refines the framework by recommending that Type 3 (Exam Literacy) be split into a knowledge subcomponent (3a: mark-scheme vocabulary, command-word decoding) and a procedural-fluency subcomponent (3b: timed performance, question-selection under pressure), and that Type 5 (Motivational) be split into an internal subcomponent (5a: the self-efficacy loop) and a circumstantial subcomponent (5b: absence of study time, competing demands). The companion system paper (Khondokar, 2026) already operationalises these splits through adaptive assessment (untimed vs timed comparison for 3a/3b) and engagement analytics (pattern-based detection for 5a/5b). These splits are adopted in the system design before being formally tested in the conceptual framework; the first research cycle will determine whether the empirical data supports them.

7 Conclusion

This paper has proposed a conceptual framework that addresses a gap in the literature on Bangladesh English-medium education by combining it with the educational psychology, cognitive science, and evolutionary psychology traditions. The framework is organised around two research questions concerning the sources of student struggle and the factors distinguishing one-sitting completers from chronic retakers in the IGCSE/IAL context; and the operationalisation of self-regulated learning, deliberate practice, cognitive engagement, cognitive load, and exam literacy as observable behavioural and outcome signatures.

The framework is grounded in a four-layer model of academic performance, Content Knowledge $\times$ Exam Literacy $\times$ Learning Strategy $\times$ Self-Regulation, that synthesises work on self-regulated learning (Education Endowment Foundation, 2018; Ergen and Kanadli, 2017; Pintrich, 2000; Zimmerman, 2002), mindset (Dweck, 2006), cognitive engagement (Chi and Wylie, 2014), self-efficacy (Bandura, 1997), grit (Duckworth et al., 2007), mastery learning (Bloom, 1968), formative assessment (Black and Wiliam, 1998; Hattie and Timperley, 2007), deliberate practice (Ericsson, 2006), spacing effects (Cepeda et al., 2006; Ebbinghaus, 1913; Pashler et al., 2007), the testing effect (Roediger and Karpicke, 2006), study strategies (Dunlosky et al., 2013), desirable difficulties (Bjork and Bjork, 2011), cognitive load theory (Sweller, 1988; Sweller et al., 2011), working memory (Cowan, 2014), and evolutionary educational psychology (Geary, 2008; Plomin and von Stumm, 2018). The six-type struggle typology and the pre-registered predictions are testable by any researcher with access to interviews, school records, or a generic learning platform.

The contribution of this paper is theoretical and pre-registrational. It offers a four-layer model of academic performance, a six-type struggle typology, and a hypothesised mechanism for the retake paradox that are testable against data that any sufficiently instrumented research setting can provide. A companion paper (Khondokar, 2026) describes one specific system designed to operationalise the framework at scale. The first cycle of the programme is currently in the design stage, and subsequent papers will report findings as data accumulate.

Answering the research questions. Returning to the research questions posed in Section 2: **RQ1** (sources of student struggle and factors distinguishing one-sitting completers from chronic retakers) is addressed by the six-type typology and the six pre-registered predictions, each of which operationalises a hypothesised distinguishing factor. The answer RQ1 receives in this paper is provisional and pre-registrational: the typology identifies the candidate sources and the predictions specify how each source should manifest behaviourally, but the empirical confirmation of which sources actually distinguish the two groups is deferred to Cycle 1. **RQ2** (operationalising SRL, deliberate practice, cognitive engagement, cognitive load, and exam literacy as observable signatures) is addressed by the derivation table (Table 2) and by the pre-prediction operationalisations in Section 5, which specify the observable signature for each construct in terms testable through interviews, school records, or learning-platform telemetry.

What this paper does not claim. This paper does not claim empirical confirmation of the framework. It does not claim that the four-layer model is the only valid organising frame for IGCSE/IAL performance, nor that the six-type typology is exhaustive. It does not claim that the framework generalises beyond the Bangladesh English-medium context without empirical test. It does not claim that heritability of cognitive capacity is the cause of the within-population

variance; the framework is agnostic on this question. It does not claim that the companion system (Paper B) is the only or the best operationalisation of the framework. Each of these claims is deferred to subsequent empirical work.

Conflict of Interest

I am the designer of a learning management system described in a companion paper (Khondokar, 2026). The framework proposed in this paper is independent of that system's implementation, but the system is designed to operationalise the framework. The conflict is managed by the separation of framework development and system evaluation functions within the research team.

Funding

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Data Availability

The framework proposed in this paper is conceptual. Empirical data from the first research cycle will be made available through a public repository upon completion of data collection.

Open Practices

The six pre-registered predictions, the analysis plan, and the analysis code stubs are deposited on the Open Science Framework at <https://osf.io/igcse-ial-four-layer> (anonymised for peer review; the link will be live upon acceptance), time-stamped prior to data collection. Upon completion of Cycle 1, the following will be released openly: (a) anonymised interview transcripts and coding schemes; (b) anonymised school-record and platform-telemetry data, with student and teacher identifiers removed and with any remaining personally identifiable information redacted; (c) the analysis code (R and Python) used to produce the results reported in the Cycle 1 results paper; (d) the interview protocol, the diagnostic-test items, and the qualitative-coding codebook. The open-materials release is intended to allow independent replication of the typology test in other high-stakes-examination contexts (IB, AP, Gaokao, JEE).

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